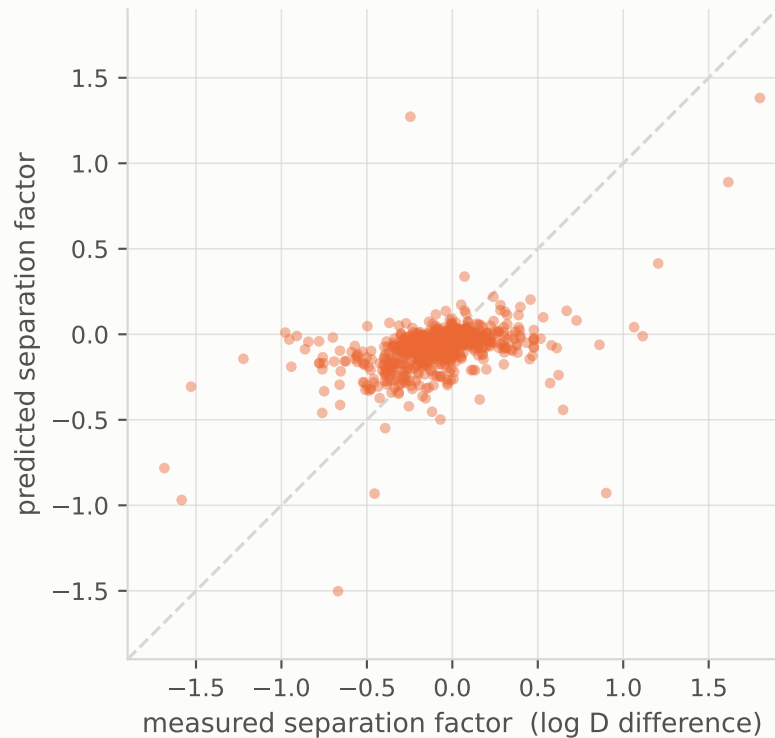


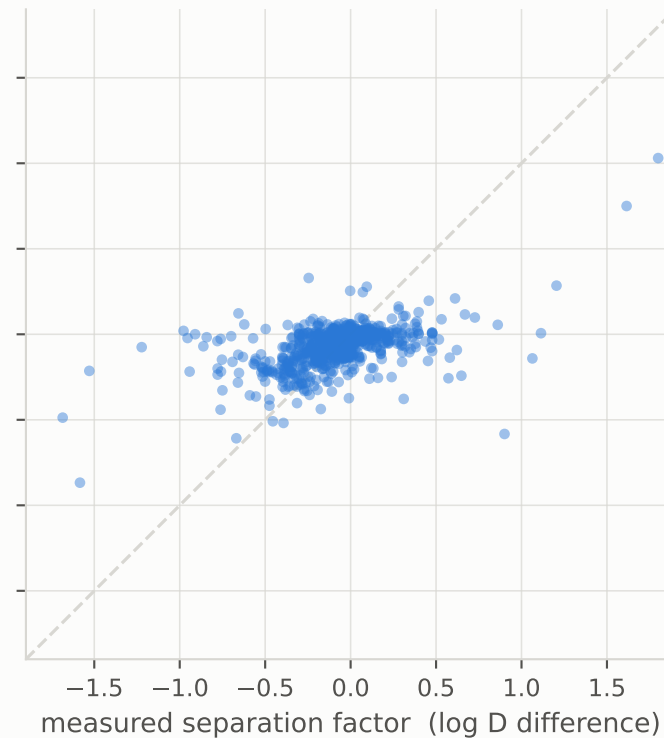
Without the 3D model

$R^2 = +0.226$



With the 3D model

$R^2 = +0.267$



905 neighbouring-lanthanide pairs, out-of-fold. Dashed line = perfect prediction. Both panels compress toward zero — the models get direction and ranking better than magnitude — but the 3D model tightens the relationship.